Local-Unitary Rigidity of Stabilizer AME States and Transversal Clifford Groups of MDS–CSS Codes

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Abstract

Let $q = p^e$. For every $m \geq 2$, each product-unitary intertwiner between two stabilizer AME$(2m, q)$ states is Clifford factor by factor. On any $m+1$ parties, the supported stabilizer labels form a q^2-element group and project bijectively onto the complete local Weyl basis at every retained party. The reduced operator therefore determines its local Weyl axes intrinsically. Consequently, every tensor-product conversion between the associated $[[2m-1, 1, m]]_q$ stabilizer quantum-MDS encoders is Clifford on every physical factor and on the logical factor. Moreover, LU equivalence of stabilizer AME states is exactly symplectic equivalence of their minimum-support operator-pushing atlases. The fixed-party projective symmetry group is an extension of the compatible atlas-gauge group by the projective stabilizer, and in prime dimension the gauge group is a holonomy centralizer in $\mathrm{SL}_2(q)$.

For equal-phase CSS states of linear $[2m, m, m+1]_q$ MDS codes, the code geometry further determines the exact transversal group. For odd prime q, the exact fixed-party projective transversal logical group is $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ precisely when C is diagonally isodual, and $\mathbb{F}_q^2 \rtimes T$ otherwise, where T is the split torus; generalized Reed–Solomon codes lie on the first branch.

For $m = 3$, a degree-eight quotient z classifies projective and monomial-code equivalence on an admitted non-GRS pencil of six-point arcs; over odd prime fields it also classifies local-Clifford and local-unitary equivalence. In that prime-field scope, with any party taken as the input of a $[[5, 1, 3]]_q$ encoder, the fixed-party logical symplectic image is $\mathrm{SL}_2(q)$ on the conic/GRS locus and the split torus off it. Operator-valued marginals recover the local Weyl frame and force rigidity, whereas every fixed-copy scalar contraction is generically constant; exact marginal and four-copy witnesses still detect special strata.

1 Introduction

An absolutely maximally entangled state has maximally mixed reductions on every subsystem no larger than half the parties. Fix a prime power $q = p^e$ and the standard additive q-dimensional Weyl system. We ask whether a stabilizer AME$(2m, q)$ state retains its local Weyl frames under arbitrary single-party changes of basis.

The local-unitary versus local-Clifford problem is false for general stabilizer states [JCWY10]. Maximal entanglement supplies the missing rigidity.

Throughout, *transversal* means a tensor product of one-site physical operators with the party labels fixed. A *fixed-party* symmetry has this form; a *party-moving* symmetry additionally permutes tensor factors. LU and LC equivalence allow a party permutation when it is displayed explicitly. Logical groups called fixed-party below exclude such permutations.

Theorem 1.1 (Local-unitary rigidity)**.** Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer AME$(2m, q)$ states for the standard additive Weyl system. Suppose that, for a party permutation π, single-qudit unitaries $U_1, \ldots, U_{2m}$, and a phase θ,

$$(U_1 \otimes \cdots \otimes U_{2m})\pi|\psi\rangle = e^{i\theta}|\phi\rangle.$$

Then every U_i normalizes the finite-field Weyl system. In particular, the displayed equivalence is local Clifford.

The mechanism is visible in any $m+1$-party reduced operator. Stabilizer labels supported on those parties form a group of order q^2, and projection to the local Pauli-label group at every retained party is bijective. The marginal is therefore diagonal on the complete local Weyl basis, with nonzero coefficients. A rank-one contraction recovers exactly one Weyl axis, so the tensor intrinsically determines every local Weyl frame. Product conjugation must permute those frames and is consequently Clifford. The argument uses neither CSS form, equal phases, classical linearity, nor $\mathbb{F}_q$-linearity of the stabilizer labels.

Corollary 1.2 (Transversal Clifford no-go)**.** Let $V_\psi, V_\phi : \mathbb{C}^q \to (\mathbb{C}^q)^{\otimes(2m-1)}$ be encoding isometries whose normalized Choi states are stabilizer AME$(2m, q)$ states $|\psi\rangle, |\phi\rangle$. They encode the associated $[[2m-1, 1, m]]_q$ quantum MDS codes. If

$$(U_1 \otimes \cdots \otimes U_{2m-1})V_\psi = V_\phi L$$

for a logical unitary L, then L and every U_i are Clifford. Thus every transversal conversion between two codes in the family is Clifford factor by factor. Taking $V_\psi = V_\phi$, no such code admits a transversal non-Clifford logical unitary.

Theorem 1.3 (Minimum-support atlas classification)**.** After a fixed party relabelling, two stabilizer AME$(2m, q)$ states are LU equivalent exactly when their minimum-support operator-pushing atlases are equivalent by local symplectic changes of Weyl frame. For one state, let $\mathcal{G}_\psi$ be its group of compatible local symplectic gauges, let Γ_ψ be its fixed-party projective product-unitary symmetry group, and let L_ψ be its projective stabilizer. Then

$$1 \longrightarrow L_\psi \longrightarrow \Gamma_\psi \longrightarrow \mathcal{G}_\psi \longrightarrow 1$$

is exact. If q is prime, choosing base transports identifies $\mathcal{G}_\psi$ with the centralizer in $\mathrm{SL}_2(q)$ of the atlas holonomies.

Rains recovered the three Pauli axes from a diagonal correlation tensor, and Van den Nest, Dehaene, and De Moor used this mechanism in a minimal-support LU-to-LC criterion for qubit stabilizer states [Rai99, Theorem 13] [VdNDDM05b, Theorem 1 and Corollary 1]. Van den Nest, Dehaene, and De Moor also gave a finite invariant set characterizing LC equivalence of arbitrary binary stabilizer states [VdNDDM05a]. Our atlas theorem is not a first general LC-classification result: it uses the AME support profile to replace a global invariant list by an all-prime-power operator-pushing criterion and an explicit symmetry-group centralizer. Tan relates the LU orbits and local symmetries of even-party AME tensors to quantum-MDS codes and computes the full local symmetry group of the canonical four-qutrit AME state [Tan26, Theorems 3.5, 3.6, and 5.3]. The local factors in Tan's displayed generators are qutrit Cliffords, so that calculation contains the $q = 3, m = 2$ automorphism subcase above. Our contributions are the uniform all-prime-power, all-m, arbitrary-additive intertwiner theorem and the exact minimum-support-atlas classification of its residual Clifford symmetries. The AME condition itself supplies both the full-Weyl marginal required by the axis argument and the minimum-support generation required by the atlas theorem.

Equivalently, the support bijection is a minimal-support Pauli form of perfect-tensor operator pushing: every one-party Weyl operator has a unique product-Weyl representative on any chosen m of the other parties. Perfect-tensor operator pushing and the associated $[[2m-1, 1, m]]$ encoding interpretation were introduced in the holographic-code setting by Pastawski, Yoshida, Harlow, and Preskill [PYHP15, Eq. (2.3) and Section 5.7]. Overlapping minimum supports compare the recovered local Weyl-label frames. Theorem 1.3 makes their transition atlas a complete classification reduction rather than only an obstruction. For six parties, explicit holonomy conjugacy data recover the pencil coordinate z. The boundary $m \geq 2$ is sharp: a Bell pair has non-Clifford $U \otimes \overline{U}$ symmetries.

Rigidity determines which product operations are possible; the MDS–CSS model determines which Clifford operations actually occur. If $C \leq \mathbb{F}_q^{2m}$ is a linear $[2m, m, m+1]_q$ MDS code, then

$$|\Psi_C\rangle = q^{-m/2} \sum_{c \in C} |c\rangle \tag{1.1}$$

is a stabilizer AME$(2m, q)$ state. Conversely, an equal-phase state of a linear $[2m, m]_q$ code is AME only if the code is MDS [RGRA18]. Here the MDS–CSS geometry is no longer needed for rigidity; it makes the exact transversal group computable.

Corollary 1.4 (Diagonal-isodual transversal logical group)**.** Let q be an odd prime, let $m \geq 2$, and let $C \leq \mathbb{F}_q^{2m}$ be a linear $[2m, m, m+1]_q$ MDS code. The projective group of logical unitaries of the associated $[[2m-1, 1, m]]_q$ quantum MDS code that admit tensor-product physical implementations is

$$\begin{cases} \mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q), & \text{if } SC = C^\perp \text{ for a nonsingular diagonal } S, \\ \mathbb{F}_q^2 \rtimes T, & \text{otherwise,} \end{cases}$$

where $T = \{\mathrm{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}$.

Put

$$\mathcal{D}(C, C^\perp) = \{s \in \mathbb{F}_q^{2m} : \mathrm{diag}(s)C \subseteq C^\perp\}.$$

The phase condition is the nullity test $\dim \mathcal{D}(C, C^\perp) = 1$. Proposition 3.8 proves that this dimension is zero or one and that every nonzero vector in the space has full support and is unique up to scalar.

The phrase "modulo phase" is essential. Over an odd field the $\mathrm{SL}_2(q)$ factor on the diagonally isodual branch has coherent Weil lifts, so no metaplectic obstruction remains there. The full affine group does not lift through the one-qudit scalar quotient: its translation subgroup is the projective Weyl group, whose two coordinate generators have a nontrivial scalar commutator. This Heisenberg obstruction is independent of the factor set for party-moving symmetries in Corollary 3.6. Generalized and extended generalized Reed–Solomon codes are diagonally isodual, but the condition is not confined to that family: self-dual non-GRS MDS constructions are known [ZL24].

For example, an extended $[8, 4, 5]_7$ Reed–Solomon code gives an AME$(8, 7)$ tensor and a $[[7, 1, 4]]_7$ quantum MDS code whose projective transversal logical group has order $7^2|\mathrm{SL}_2(7)| = 16464$. No non-Clifford transversal gate enlarges this group.

For binary stabilizer codes, Wirthmüller determines the identity component of the local-unitary automorphism group and gives a protected-qubit criterion for finiteness of the induced automorphism group [Wir11, Theorem 7 and Corollary 11]. Our result instead determines every local factor and gives projective finiteness uniformly inside the prime-power stabilizer-AME class. The stabilizer qualifier is essential: arbitrary phased minimal-support AME$(6, d)$ states form infinitely many LU classes [RL25]; nonlinear orthogonal arrays and nonstabilizer AME states are not covered.

The geometric applications in the remainder of the paper specialize to $m = 3$. Our main application is the ordered pencil

$$H_t = \big((0, 1, 1-t), (0, 1, t-1), (1, 1-t, 0), (1, t-1, 0), (1, 0, -t), (1, 0, t)\big). \tag{1.2}$$

On its admitted odd-prime-field non-GRS locus, define

$$\begin{aligned} A(t) &= -4t(t-1)^2, \\ B(t) &= (t^2-t+1)(t^2-3t+1), \\ z(t) &= \big(B(t)/A(t)\big)^2. \end{aligned} \tag{1.3}$$

Exact bracket identities show that z is the degree-eight projective quotient of the pencil. Theorem 1.1 upgrades its local-Clifford classification to the following complete local result.

Corollary 1.5 (Classification of the admitted pencil)**.** For admitted parameters t, u over a prime field $\mathbb{F}_q$ with q odd, the following are equivalent:

$$H_t \sim_{\text{proj}} H_u \iff \ker H_t \sim_{\text{mon}} \ker H_u \iff \Psi_t \sim_{\text{LC}} \Psi_u \iff \Psi_t \sim_{\text{LU}} \Psi_u \iff z(t) = z(u),$$

where party permutations are allowed.

The exact diagonal-isodual group dichotomy above holds for every m. For six parties it has a geometric coding-theoretic interpretation over odd prime fields. Regard any one tensor leg as the input of the associated $[[5,1,3]]_q$ encoder. The fixed-party product-Clifford symmetries induce the full logical $\mathrm{SL}_2(q)$ precisely when the six-arc is conic/GRS; otherwise they induce only the split torus. Thus the nineteenth-century self-association condition for six points controls which one-qudit logical Clifford gates are available without moving parties. Including logical Paulis, the exact fixed-party projective logical group is $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ on the GRS locus and $\mathbb{F}_q^2 \rtimes T$ off it.

Quantum Reed–Solomon codes and their encoding circuits were developed by Grassl, Geiselmann, and Beth [GGB99]. Aharonov and Ben-Or constructed fault-tolerant coordinatewise gates for polynomial codes, with degree reduction when a gate changes the polynomial degree [ABO08, Section 5]; Gottesman constructed the full logical Clifford group for any prime-dimensional stabilizer code using ancillas and Pauli measurements [Got99]. These results establish that GRS and polynomial encoders support fault-tolerant logical gates, but they do not identify the fixed-party product-Clifford kernel. Corollary 1.4 combines diagonal code–dual equivalence with LU rigidity to identify the exact transversal logical group for every half-dimensional MDS code. In coding theory, diagonal code–dual equivalence is the fixed-coordinate form of isometry duality; it occurs in algebraic-geometry MDS families as well as GRS codes [ZZ26]. For binary self-dual CSS codes, Tansuwannont, Takada, and Fujii give criteria for transversal Hadamard and phase operations relative to a compatible logical basis [TTF25]. Their setting and conclusion differ from the odd-prime, one-logical-qudit exact fixed-party group here. Anderson and Jochym-O'Connor constrain diagonal transversal gates, including maps between two qubit stabilizer codes, to Clifford-hierarchy angles [AJO14, Propositions 3 and 4]. Sayginel et al. derive SWAP-transversal logical Cliffords from code automorphisms, including Pauli phase correction and logical-action recovery [SKW+25]. Our conversion theorem instead proves that every factor of an arbitrary product-unitary conversion between stabilizer quantum-MDS encoders in Corollary 1.2 is Clifford; the MDS–CSS specialization then determines exact extensions and factor sets rather than an automorphism-search procedure. For qubit stabilizer codes, Dasu and Burton classify diagonal transversal Clifford gates on any number of identical code blocks by the code's $\mathbb{F}_2$-endomorphism algebra [DB25, Theorems 5.5 and 6.1]. Their classification permits gates coupling corresponding qubits across several blocks but applies the same local tableau at every physical coordinate. We instead allow an independent one-qudit factor at each coordinate, work over odd prime fields, and determine the exact one-block logical image for the half-dimensional MDS/CSS family. The transition atlas gives the two settings a common algebraic form: compatible local blocks centralize loop transports, while Dasu–Burton's gates are selected by a code endomorphism algebra. The locality and block hypotheses remain complementary. Recent high-rate constructions obtain the full logical Clifford group by allowing fold-transversal or 2-local transversal layers [TCT26, Hol26]. Those architectures encode many logical qubits and deliberately use interactions beyond a single fixed-party product layer. Conversely, Chakraborty and Gottesman prove that a stabilizer code encoding more than one logical qubit cannot realize its full logical Clifford group by fully transversal gates [CG26]. The exact positive result here concerns one logical qudit, so it lies on the permitted side of that obstruction and determines which half-dimensional MDS encoders attain the full one-qudit group. Theorem 5.2 identifies the geometric condition among six-arcs: self-association is exactly the condition for the fixed-party logical symplectic image to exceed the split torus. The comparison is with isometry-dual MDS constructions, quantum Reed–Solomon and polynomial-code gate constructions, self-dual CSS transversal Clifford gates, general prime-dimensional stabilizer Clifford synthesis, and restrictions on transversal gate sets. We claim the exact group statements, not priority for the underlying code or gate constructions. Eastin and Knill's restriction on universal transversal sets gives the general boundary [EK09].

Classically, we use Coble's association theory [Cob22] through the modern treatment of Dolgachev and Ortland [DO88, Chapter III]; for finite-projective arc background see Segre, Hirschfeld, and the modern survey of Ball and Lavrauw [Seg55, Hir98, BL19]. Over characteristic zero, the GIT quotient of six ordered points in $\mathbb{P}^2$ is a double cover of $\mathbb{P}^4$, with Gale association exchanging the sheets and the self-associated locus forming the Igusa-quartic branch divisor [HMSV08, Section 2.3]. That divisor is dual to the Segre cubic parametrizing six points on $\mathbb{P}^1$. Our six-arc phase theorem is the finite-field, fixed-label slice of this picture: the branch condition becomes diagonal isoduality, and the Veronese determinant detects the conic/GRS locus. This interpretation is not used in the proof.

The rest of the paper separates classification from explicit scalar certificates. Section 2 fixes the arc, code, Weyl, and AME conventions. Section 3 proves the headline theorem and its transversal and GRS corollaries. Section 4 proves the z-classification, and Section 5 proves the logical-Clifford phase theorem: $\mathrm{SL}_2(q)$ on the GRS locus and a split torus off it. Section 6 gives a uniform H_3 (icosahedral)/GRS marginal separator, an exact four-copy $q = 13$ witness, and the generic-constancy obstruction for fixed-copy scalars. Appendix A calculates the exceptional divisor of that witness, redundant given Corollary 1.5; it is not part of the classification. Section 7 records the computational trust boundary and the precise scope of all results.

2 Stabilizer AME states and MDS–CSS models

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i\ \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x + a\rangle, \qquad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

For n parties, a pure stabilizer state is fixed by an abelian Pauli group $\mathcal{S}$ of order q^n. Its projective label group $L \le (\mathbb{F}_q^2)^n$, viewed additively over $\mathbb{F}_p$, is maximal isotropic for the trace-symplectic form. This is the standard arbitrary-additive prime-power stabilizer correspondence [KKKS06, Theorems 13 and 15]. For each $\ell \in L$, write the stabilizer lift as $\widetilde{W}_\ell = \alpha_\ell W_\ell$, where $\alpha_\ell \neq 0$. Then

$$|\psi\rangle\langle\psi| = q^{-n} \sum_{\ell \in L} \widetilde{W}_\ell. \tag{2.1}$$

If A is a party set, put

$$L(A) = \{\ell \in L : \operatorname{supp}(\ell) \subseteq A\}.$$

Weyl orthogonality and partial trace give

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L(A)} \widetilde{W}_\ell. \tag{2.2}$$

In particular, ρ_A^ψ is maximally mixed exactly when $L(A) = \{0\}$. Thus a stabilizer state on $2m$ parties is AME exactly when it has no nonidentity stabilizer label supported on at most m parties.

The rigidity theorem uses only (2.1)–(2.2). For exact group and classification results, we now specialize to the six-party MDS–CSS model.

A six-arc is an ordered sextuple of points of $\mathbb{P}^2(\mathbb{F}_q)$ with no three collinear. Choose nonzero column representatives and collect them in a 3-by-6 matrix H. The arc condition says that every three-column minor of H is nonzero. Consequently

$$C = \ker H \leq \mathbb{F}_q^6$$

is an $[6, 3, 4]_q$ MDS code, and its dual $C^\perp = \operatorname{rowspace} H$ is MDS with the same parameters. Conversely, a parity-check matrix of such a code supplies a six-arc. Projectivities, independent rescaling of columns, and permutation of columns correspond exactly to monomial equivalences of the kernels. These objects exist exactly when $q \geq 4$. Indeed, from one point of a six-arc the five joining lines are distinct, so $q + 1 \geq 5$. For $q = 4$, a conic together with its nucleus is a six-arc; for $q \geq 5$, any six points of a nonsingular conic form a six-arc.

For the resulting CSS stabilizer we use the exact lift

$$W_{(c,h)} = X(c)Z(h) = \bigotimes_{i=1}^{6} X(c_i)Z(h_i), \qquad (c, h) \in C \oplus C^\perp.$$

Since $h \cdot c' = 0$ for $h \in C^\perp$ and $c' \in C$, these lifts multiply without a residual phase and fix $|\Psi_C\rangle$.

Proposition 2.1 (Arc–AME dictionary)**.** For a six-arc with kernel C, the state $|\Psi_C\rangle$ in (1.1) is a minimal-support stabilizer AME$(6, q)$ state. Its projective Pauli-label Lagrangian is

$$L_C = (C \oplus 0) + (0 \oplus C^\perp) \leq \mathbb{F}_q^{12}.$$

Monomial code equivalence, together with a party permutation, induces local-Clifford equivalence of the corresponding states.

Proof. The operators $X(c)$, $c \in C$, fix the equal superposition. The operators $Z(h)$, $h \in C^\perp$, fix it because $\chi(h \cdot c) = 1$. The two subgroups commute, and their label space has dimension six and is Lagrangian.

For any three-coordinate set I, puncturing C to I is bijective: injectivity follows from distance four and both spaces have dimension three. In the partial trace of $|\Psi_C\rangle\langle\Psi_C|$, equality on the complementary three coordinates therefore forces equality of the codewords, while the projection to I runs once through $\mathbb{F}_q^3$. Hence every three-party reduction is $q^{-3}I$. Smaller reductions are maximally mixed by tracing again, so the state is AME$(6, q)$. Its support has the minimum possible size q^3.

A column multiplier $x \mapsto ax$ is a single-party Clifford operation: it sends $X(b)$ to $X(ab)$ and $Z(b)$ to $Z(a^{-1}b)$. Projective row operations do not change the kernel, and column permutations permute parties. This proves the last assertion. $\square$

For a party set S, let

$$L_C(S) = \{\ell \in L_C : \operatorname{supp}(\ell) \subseteq S\}.$$

The general stabilizer partial-trace formula specializes to

$$\rho_S^C = q^{-|S|} \sum_{\ell \in L_C(S)} W_\ell. \tag{2.3}$$

For later block calculations, write

$$C_X = C \oplus 0, \qquad C_Z^\perp = 0 \oplus C^\perp, \qquad L_C = C_X \oplus C_Z^\perp.$$

This formula is the bridge used in both the rigidity proof and the marginal invariants below. A Pauli-labelled quantity is not by itself an LU invariant; the invariant statements will always be expressed either as covariance of the reduced operators or as basis-free tensor contractions.

The GRS boundary has a simple geometric meaning. A dimension-three length-six generalized Reed–Solomon code is represented, up to column multipliers, by six points on a nonsingular conic in $\mathbb{P}^2$. We use "GRS state" for the state of any such code. The non-GRS pencil in Section 4 consists of arc matrices H_t away from that conic locus and from the other factors needed to keep the columns distinct and the displayed coordinates defined.

3 Full-Weyl marginals and LU rigidity

We first isolate the linear-algebra statement that recovers a diagonal tensor's coordinate axes. It is the full-Weyl counterpart of the three-Pauli argument in [Rai99, VdNDDM05b].

Lemma 3.1 (Diagonal-tensor axes)**.** Let $r \geq 3$, let $E_1, \ldots, E_r$ be N-dimensional complex inner product spaces, and choose orthonormal bases $\{e_{ij} : 1 \leq j \leq N\}$. If

$$T = \sum_{j=1}^{N} \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \qquad \lambda_j \neq 0, \tag{3.1}$$

then the coordinate axes in every E_i are intrinsic to T. Consequently, a product of invertible linear maps carrying one tensor of the form (3.1) to another is monomial in the displayed bases.

Proof. Contract the first factor against $x = \sum_j x_j e_{1j}^*$. The result is

$$T_x = \sum_j \lambda_j x_j e_{2j} \otimes \cdots \otimes e_{rj}.$$

Flattening from E_2 to $E_3 \otimes \cdots \otimes E_r$ gives a matrix of rank $\#\{j : x_j \neq 0\}$. Thus T_x has rank one exactly when x lies on a coordinate axis. Product equivalence preserves this rank-one contraction locus, and hence permutes the axes in the first factor. Apply the same argument in every factor. □

Call an operator on $r \geq 3$ qudits *full-Weyl diagonal* if there is a q^2-element set P, a distinguished element $0 \in P$, and bijections $f_i : P \to \mathbb{F}_q^2$ with $f_i(0) = 0$, for which

$$R = \sum_{v \in P} \lambda_v \bigotimes_{i=1}^{r} W_{f_i(v)}, \qquad \lambda_v \neq 0. \tag{3.2}$$

The coefficients need not be equal. What matters is that the same q^2-element index set runs bijectively through the complete local Weyl basis at every party. Neither P nor the bijections need be $\mathbb{F}_q$-linear.

Proposition 3.2 (Full-Weyl marginal criterion)**.** Let R and R' be full-Weyl diagonal operators on the same $r \geq 3$ parties, possibly with different indexing sets, bijections, and coefficients. If

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger,$$

then every U_i is Clifford.

Proof. View R and R' as tensors in the local Hilbert–Schmidt spaces. Both have the form in Lemma 3.1, with $N = q^2$. Local conjugation acts invertibly on each factor, so it must permute the Weyl axes. It fixes the identity operator and hence the identity axis. It therefore permutes the nonidentity projective Weyl operators, which is exactly the single-qudit Clifford condition. □

Corollary 3.3 (Full-Weyl marginal-cover rigidity)**.** Let $|\psi\rangle, |\phi\rangle$ be n-qudit states and suppose a product unitary, after a party relabelling, carries $|\psi\rangle$ to $|\phi\rangle$ up to phase. If every party belongs to a set S of at least three parties for which both reduced operators ρ_S^ψ and ρ_S^ϕ are full-Weyl diagonal, then every local factor is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each chosen S. Apply Proposition 3.2; the chosen sets cover all parties. □

Proposition 3.4 (Stabilizer-AME support theorem)**.** Let $|\psi\rangle$ be a stabilizer AME$(2m, q)$ state, and let A be any $m + 1$ parties. Then

$$|L(A)| = q^2.$$

Every nonidentity label in $L(A)$ has support exactly A, and for each $i \in A$ the coordinate projection

$$p_i : L(A) \longrightarrow \mathbb{F}_q^2$$

is bijective.

Proof. The projective label group L has order q^{2m}. Restrict its labels to the $m - 1$ parties outside A. The target has order $q^{2(m-1)}$, and the kernel is $L(A)$; hence

$$|L(A)| \geq q^{2m}/q^{2(m-1)} = q^2. \tag{3.3}$$

Fix $i \in A$. A label in the kernel of p_i is supported on the m-set $A \setminus \{i\}$. The AME condition and (2.2) exclude every such nonidentity stabilizer, so p_i is injective. Its codomain has order q^2, giving the reverse inequality in (3.3) and hence bijectivity. The same no-small-support argument shows that every nonidentity element of $L(A)$ has full support A. □

Proposition 3.5 (A stabilizer-AME marginal fixes the local Weyl axes)**.** Let $|\psi\rangle, |\phi\rangle$ be stabilizer AME$(2m, q)$ states. If a product unitary on an $(m+1)$-set A carries ρ_A^ψ to ρ_A^ϕ, then every local factor on A is Clifford.

Proof. By (2.2),

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L_\psi(A)} \widetilde{W}_\ell.$$

Index this sum by $P = L_\psi(A)$. Proposition 3.4 says that every coordinate projection $P \to \mathbb{F}_q^2$ is bijective. Relative to the fixed Weyl convention, each stabilizer lift contributes a nonzero coefficient. Thus ρ_A^ψ is full-Weyl diagonal in the sense of (3.2), and the same holds for ρ_A^ϕ. Apply Proposition 3.2. □

Proof of Theorem 1.1. Absorb the party permutation by relabelling $|\phi\rangle$. Partial trace gives a product-unitary equivalence $\rho_A^\psi \sim \rho_A^\phi$ for every $(m+1)$-set A. Given a party i, choose such a set containing it and apply Proposition 3.5. Thus every U_i is Clifford. □

The same support calculation recovers standard quantum-MDS structure. If $|A| \le m$, the AME condition gives $L(A) = \{0\}$. If $|A| > m$, then $\rho_{A^c}^\psi$ is maximally mixed, so purity of the global state gives $\mathrm{Tr}((\rho_A^\psi)^2) = q^{-(2m-|A|)}$. On the other hand, (2.2) and Weyl orthogonality give $\mathrm{Tr}((\rho_A^\psi)^2) = q^{-|A|}|L(A)|$. Hence, for every party set A,

$$|L(A)| = q^{2\max(|A|-m,0)}. \tag{3.4}$$

Möbius inversion is the stabilizer-state specialization of the quantum-MDS Shor–Laflamme distribution of Huber and Grassl [HG20, Theorems 8 and 9]; in particular the number of weight-$m+1$ projective stabilizers is $\binom{2m}{m+1}(q^2-1)$. We use one further consequence. If $|A| \ge m+2$ and $i \ne j$ lie in A, then the dimensions in (3.4), together with

$$L(A \setminus \{i\}) \cap L(A \setminus \{j\}) = L(A \setminus \{i,j\}),$$

give

$$L(A) = L(A \setminus \{i\}) + L(A \setminus \{j\}). \tag{3.5}$$

Descending induction shows that the $(m+1)$-supported subgroups span the full label group. This is the additive quantum-MDS form of minimum-weight generation, not a separate weight-enumerator claim.

We now prove the atlas classification stated in the introduction. Regard each local label group

$$P_i = \mathbb{F}_q^2$$

as a $2e$-dimensional trace-symplectic space over $\mathbb{F}_p$. For every $(m+1)$-set A and $i, j \in A$, the support theorem defines the operator-pushing transition

$$M_A(i,j) = p_j p_i^{-1} : P_i \longrightarrow P_j.$$

For two states ψ, ϕ on the same labelled parties, a *symplectic atlas equivalence* is a tuple $F_i \in \mathrm{Sp}(P_i)$ satisfying

$$F_j M_A^\psi(i,j) = M_A^\phi(i,j) F_i \quad \text{for every } A \text{ and } i,j \in A. \tag{3.6}$$

Here $\mathrm{Sp}(P_i) = \mathrm{Sp}_{2e}(\mathbb{F}_p)$ is the action induced by the standard additive local Clifford group.

Proof of Theorem 1.3. First fix the party labels. A block symplectic map $F = \bigoplus_i F_i$ carrying L_ψ to L_ϕ carries every supported subgroup $L_\psi(A)$ to $L_\phi(A)$. Writing these subgroups as graphs through any projection p_i gives exactly (3.6).

Conversely, suppose (3.6) holds. Fix A and $i \in A$. Every element of $L_\psi(A)$ is uniquely determined by its i-coordinate, and the equations with $j \in A$ show that its image under F is the unique element of $L_\phi(A)$ with i-coordinate $F_i p_i(\ell)$. Thus $FL_\psi(A) = L_\phi(A)$. Since the $(m+1)$-supported subgroups generate the full label groups, $FL_\psi = L_\phi$.

Choose local Clifford lifts of the F_i. Their product carries the stabilizer axes of ψ to those of ϕ, possibly with a different stabilizer character. The trace-symplectic pairing identifies the ambient label space modulo $L_\phi = L_\phi^\perp$ with L_ϕ^*, so a product Pauli cancels that character. Hence an atlas equivalence lifts to an LC equivalence. The reverse implication follows from the induced label action, and Theorem 1.1 replaces LC by LU. A party permutation is absorbed by relabelling first.

For automorphisms, the induced-label map $\Gamma_\psi \to \mathcal{G}_\psi$ is surjective by the same lifting and phase-correction argument. Its kernel consists of projective product Paulis that fix the state, namely the labels in $L_\psi^\perp = L_\psi$. This proves the exact sequence.

Finally suppose $q = p$. Choose a base party and, for every other party, one atlas transition from the base as transport. Compatibility propagates every local block from its base value. Each remaining transition is equivalent to commutation of that base block with the corresponding loop holonomy. Since $\mathrm{Sp}_2(\mathbb{F}_q) = \mathrm{SL}_2(q)$ is normal in $\mathrm{GL}_2(q)$, all propagated blocks remain symplectic. Evaluation at the base therefore identifies $\mathcal{G}_\psi$ with the holonomy centralizer inside $\mathrm{SL}_2(q)$. $\square$

Let $C \le \mathbb{F}_q^{2m}$ be $[2m, m, m+1]_q$ MDS, and fix a party set S of size $m+1$. Shortening C on the $m-1$ coordinates outside S gives an $[m+1, 1, m+1]_q$ MDS code. The dual $C^\perp$ has the same parameters, so its corresponding shortening does as well. Choose nonzero shortened words $c^S \in C$ and $h^S \in C^\perp$. Each has support exactly S, and

$$L_C(S) = \{(ac^S, bh^S) : a, b \in \mathbb{F}_q\}. \tag{3.7}$$

At a party $i \in S$, projection of this plane to the local Pauli labels is

$$(a, b) \longmapsto (ac_i^S, bh_i^S),$$

which is invertible because both coordinates are nonzero. Thus all q^2 labels in $L_C(S)$ run bijectively through the full local Weyl basis at every party of S; in particular, the $q^2 - 1$ nonidentity labels run bijectively through the nonidentity Weyl labels.

Proof of Corollary 1.2. The input marginal of each AME Choi state is $q^{-1}I$, so one-leg reshaping gives an isometry. The projection of its stabilizer label group onto the input Weyl-label space is surjective: otherwise a nonzero local label orthogonal to the image would lie in the maximal isotropic space $L^\perp = L$, contradicting the absence of a one-party stabilizer. The kernel therefore has q^{2m-2} elements and stabilizes the image, making it a one-logical-qudit stabilizer code. AME corrects erasure of any $m-1$ outputs, so its distance is at least m; the quantum Singleton bound gives equality. This is the standard perfect-tensor/quantum-MDS correspondence used in [Tan26, Theorems 3.5 and 3.6].

Let $|\Phi\rangle = q^{-1/2}\sum_x |x, x\rangle$, and use the Choi convention $|\psi\rangle = (I \otimes V_\psi)|\Phi\rangle$ and $|\phi\rangle = (I \otimes V_\phi)|\Phi\rangle$. Writing $U_{\text{phys}} = \bigotimes_i U_i$, the relation $U_{\text{phys}} V_\psi = V_\phi L$ gives

$$(I \otimes U_{\text{phys}})|\psi\rangle = (L^T \otimes I)|\phi\rangle.$$

Hence

$$\big((L^T)^{-1} \otimes U_{\text{phys}}\big)|\psi\rangle = |\phi\rangle.$$

Theorem 1.1 makes every displayed local factor Clifford. Entrywise conjugation preserves the finite-field Weyl group, since $\overline{X(a)} = X(a)$ and $\overline{Z(b)} = Z(-b)$; it therefore preserves the Clifford normalizer. Taking adjoints preserves the normalizer as well, and $U^T = (\overline{U})^\dagger$. Thus transposition preserves the finite-field Clifford group, and L is Clifford. $\square$

Corollary 3.6 (Discrete product-unitary symmetry)**.** For every stabilizer AME state in Theorem 1.1, the product-unitary automorphism group modulo one-site scalar phases is finite and discrete. Its identity component before that quotient consists only of one-site scalar phases. More precisely, if $T = (S^1)^{2m}$, G is the fixed-party product-unitary automorphism group, and $\widetilde{G}$ also displays party permutations, then

$$1 \longrightarrow T \longrightarrow G \longrightarrow \Gamma \longrightarrow 1, \qquad 1 \longrightarrow T \longrightarrow \widetilde{G} \longrightarrow \widetilde{\Gamma} \longrightarrow 1$$

are short exact sequences with closed connected kernel and finite discrete quotient; both inclusions of T are continuous. In particular, G and $\widetilde{G}$ are finite unions of connected components, each a translate of T. If Π is the subgroup of party permutations realized by $\widetilde{G}$, there is a further exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \widetilde{\Gamma} \longrightarrow \Pi \longrightarrow 1.$$

This last extension splits exactly when the realized permutations admit a homomorphic choice of projective product-unitary lifts. It has a section-free outer action $\Pi \to \mathrm{Out}(\Gamma)$: conjugation by any lift of $\pi \in \Pi$ gives an automorphism of Γ, and changing the lift changes that automorphism by an inner automorphism. For a normalized set-theoretic section $s(1) = 1$, write

$$s(\pi)s(\rho) = f_s(\pi, \rho)s(\pi\rho), \qquad f_s(\pi, \rho) \in \Gamma.$$

Then $f_s(1, \pi) = f_s(\pi, 1) = 1$, and, with α_π denoting conjugation by $s(\pi)$,

$$f_s(\pi, \rho)f_s(\pi\rho, \upsilon) = \alpha_\pi(f_s(\rho, \upsilon))\, f_s(\pi, \rho\upsilon).$$

If another normalized section is $s'(\pi) = b(\pi)s(\pi)$, where $b(1) = 1$, then

$$f_{s'}(\pi, \rho) = b(\pi)\alpha_\pi(b(\rho))\, f_s(\pi, \rho)b(\pi\rho)^{-1}.$$

The order of these factors is essential because Γ need not be abelian. The extension splits exactly when this factor set can be changed to the constant identity factor set by such a normalized b.

Proof. Every local factor of an automorphism is Clifford. The projective single-qudit Clifford group over a finite field is finite, as is the party-permutation group. The kernel of projectivizing each local factor is exactly the torus of one-site scalar phases. That torus is the identity component, hence is closed; the finite Hausdorff quotient is discrete. Translation identifies every connected component with the scalar torus, so finitely many quotient representatives cover the group. Forgetting the local factors maps $\widetilde{\Gamma}$ onto the realized permutation subgroup Π. Its kernel consists exactly of the fixed-party projective automorphisms, which gives the third sequence. A splitting is, by definition, a group-homomorphic right inverse; arbitrary or generator-by-generator lifts do not provide one. Conjugation by the middle group descends to the outer action because elements of the kernel act by inner automorphisms. Associativity of three chosen lifts gives the displayed factor-set identity. Substituting $s'(\pi) = b(\pi)s(\pi)$ gives the change-of-section formula without commuting any kernel factors. Finally, the factor set is identically one precisely when the section preserves multiplication, which proves the splitting characterization. □

Corollary 3.7 (Computed party-extension splittings)**.** This finite application is not used in the uniform results below. The realized projective party-permutation extension in Corollary 3.6 splits for both non-GRS $q = 11$ pencil classes, the $q = 11$ GRS evaluation-set representatives

$$\{0, 1, 2, 3, 4, 5\},\ \{0, 1, 2, 3, 4, 6\},\ \{0, 1, 2, 3, 5, 6\},\ \{0, 1, 2, 3, 5, 9\},$$

the enhanced-symmetry $t = -1$ rows over $\mathbb{F}_{17}$ and $\mathbb{F}_{31}$, and the H_3 representatives $(q, \tau) = (11, 4), (19, 5), (29, 6), (31, 13)$. For every listed non-GRS or H_3 row, party parity acts on the fixed split torus T by inversion. Thus a realized odd party permutation enlarges the logical linear group exactly from T to

$$N(T) = \langle T, J\rangle, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I, \qquad J\,\mathrm{diag}(a, a^{-1})J^{-1} = \mathrm{diag}(a^{-1}, a).$$

Hence $N(T)/T \cong C_2$. The linear extension does not split in odd characteristic, although its projectivization does.

Proof. For each row, an exact finite computation reconstructs the fifteen four-support shortened planes, exhausts all 720 party permutations, and checks the propagated local symplectic blocks against the full CSS Lagrangian. After quotienting by the fixed-party group identified in Theorem 5.2, it is enough to test anchor block I on the GRS locus and anchor blocks I, J off it; this discards no candidate. In the order in which the six $q = 11$ rows are listed above, their party images are

$$S_5, \quad S_4, \quad V_4, \quad C_5, \quad S_3, \quad D_{12},$$

where D_{12} has order 12. The $q = 17$ and $q = 31$ enhanced rows have party images S_4 and S_5, respectively, and each listed H_3 image is S_5.

The certificate records the complete normalized factor set for each section. It also closes a complement of order $|\Pi|$ whose projection to Π is bijective, and verifies that the associated normalized cochain changes the factor set to the identity. Hence every listed extension splits. Conjugating the fixed torus by the stored lifts gives the sign action; any odd lift therefore induces the nontrivial class of $N(T)/T$. The complement is an involution in the projective party-extension group; its determinant-one logical representative is J, whose square is $-I$. $\square$

Proposition 3.8 (The diagonal-multiplier line)**.** Let $E, F \leq \mathbb{F}_q^{2m}$ be linear $[2m, m, m+1]_q$ MDS codes, and put

$$\mathcal{D}(E, F) = \{s \in \mathbb{F}_q^{2m} : \operatorname{diag}(s)E \subseteq F\}.$$

Every nonzero $s \in \mathcal{D}(E, F)$ has full coordinate support and $\operatorname{diag}(s)E = F$. Consequently

$$\dim_{\mathbb{F}_q} \mathcal{D}(E, F) \leq 1, \qquad \mathcal{D}(E, E) = \mathbb{F}_q(1, \ldots, 1).$$

In particular, $\mathcal{D}(C, C^{\perp})$ has dimension zero or one, and C is diagonally isodual exactly in the dimension-one case. When that line exists, any two nonzero witnesses differ by a unique nonzero scalar, so every ratio s_i/s_j is intrinsic.

Proof. Let $s \neq 0$, and write $D = \operatorname{diag}(s)$. If D had support of size at most m, choose a word of E nonzero at a supported coordinate. Its image would be a nonzero word of F of weight at most m, contrary to the distance of F. Thus D has at least $m + 1$ nonzero entries. Its kernel on E consists of words supported on at most $m - 1$ coordinates, so the distance of E makes that kernel zero. Equal dimensions now give $DE = F$. A zero diagonal entry would force every word of F to vanish at that coordinate, which an MDS code does not, so s has full support.

If two multipliers were independent, a nonzero linear combination could be chosen to vanish at one coordinate, contradicting full support. Hence $\dim \mathcal{D}(E, F) \leq 1$. The all-ones vector belongs to $\mathcal{D}(E, E)$, proving the self-multiplier statement. Taking $F = C^{\perp}$ gives the remaining assertions. $\square$

Proof of Corollary 1.4. Write a fixed-party product-Clifford action on Pauli labels as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix},$$

and let $D_\alpha, D_\beta, D_\gamma, D_\delta$ be the diagonal matrices with the corresponding entries. Preservation of $L_C = C_X \oplus C_Z^{\perp}$ is equivalent to

$$\begin{aligned} D_\alpha C \subseteq C, \quad & D_\gamma C \subseteq C^{\perp}, \\ D_\beta C^{\perp} \subseteq C, \quad & D_\delta C^{\perp} \subseteq C^{\perp}. \end{aligned} \tag{3.8}$$

If the block at the chosen input is nondiagonal, then D_β or D_γ is nonzero. Proposition 3.8 and (3.8) give

$$D_\gamma C = C^{\perp} \qquad \text{or} \qquad D_\beta C^{\perp} = C.$$

The second equality can be inverted, so either one supplies a nonsingular diagonal S with $SC = C^{\perp}$. Consequently, on the non-isodual branch every input block is diagonal. Its determinant is one, and the common blocks $\operatorname{diag}(a, a^{-1})$ always preserve L_C; hence the exact linear factor there is T.

Now suppose

$$S = \operatorname{diag}(s_1, \ldots, s_{2m}), \qquad SC = C^{\perp}. \tag{3.9}$$

The projective vector $[s_1 : \cdots : s_{2m}]$, and hence every propagation ratio s_i/s_j, depends only on C.

For $\lambda, \mu \in \mathbb{F}_q$, put

$$F_i^{-}(\lambda) = \begin{pmatrix} 1 & 0 \\ \lambda s_i & 1 \end{pmatrix}, \qquad F_i^{+}(\mu) = \begin{pmatrix} 1 & \mu s_i^{-1} \\ 0 & 1 \end{pmatrix}. \tag{3.10}$$

On the CSS label space $L_C = (C \oplus 0) + (0 \oplus C^{\perp})$, the product of the lower blocks sends

$$(c, h) \longmapsto (c, h + \lambda S c),$$

and the product of the upper blocks sends

$$(c, h) \longmapsto (c + \mu S^{-1} h, h).$$

Thus both products preserve L_C and lift to product-Clifford tensor symmetries after product-Pauli corrections of their stabilizer phases. Over the prime field, the phase lift is explicit. If a product Clifford U preserves the label Lagrangian L_C, then $U|\Psi_C\rangle$ is stabilized by the same Pauli axes, but possibly with a character $\chi : L_C \to \langle \omega \rangle$. Write $\chi(\ell) = \omega^{f(\ell)}$ for a linear functional $f \in L_C^*$. Since $L_C = L_C^{\perp}$, the symplectic pairing induces an isomorphism

$$\mathbb{F}_q^{4m}/L_C \longrightarrow L_C^*, \qquad [v] \longmapsto (\ell \mapsto \langle v, \ell \rangle).$$

Choose v mapping to $-f$. Conjugation by the product Pauli W_v cancels χ, so $W_v U$ fixes the one-dimensional joint $+1$ eigenspace of L_C, up to an overall scalar. Thus label-space preservation is sufficient here; no unproved coherent choice of Clifford phases is being used. At any chosen input party j, their blocks run through all lower and upper elementary unipotents because $s_j \neq 0$. Together these generate $\mathrm{SL}_2(q)$. Under the Choi correspondence, an input block A induces the logical block A^{-T}. This map is an automorphism of $\mathrm{SL}_2(q)$, so every one-qudit logical symplectic action has a product-Clifford physical implementation. Logical Paulis have physical Pauli representatives, so the full projective one-qudit Clifford group $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ is transversal.

Corollary 1.2 excludes every product-unitary logical action outside that group. Hence the displayed group is exact.

For a generalized or extended generalized Reed–Solomon code, the standard dual formula supplies (3.9), including the extended evaluation point at length $q + 1$. Thus the GRS tower lies on the diagonally isodual branch, but the proof used no evaluation structure.

There are two different lift questions behind this projective statement. Fix an input coordinate j, put $t_i = s_i/s_j$, and write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(q).$$

Propagation to coordinate i is

$$\phi_i(A) = \begin{pmatrix} a & t_i^{-1} b \\ t_i c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & t_i \end{pmatrix} A \begin{pmatrix} 1 & 0 \\ 0 & t_i^{-1} \end{pmatrix}. \tag{3.11}$$

Thus every ϕ_i is a homomorphism, ϕ_j is the identity, and the elementary matrices in (3.10) satisfy all $\mathrm{SL}_2(q)$ relations coordinatewise. This calculation also applies at the extended evaluation point. It uses only the diagonal duality $SC = C^{\perp}$, not the evaluation formula for a GRS code.

For odd q, the finite Stone–von Neumann representation of the Heisenberg group H_q extends to a genuine representation of $\mathrm{SL}_2(q) \ltimes H_q$ [Gér76, Theorem 1]. Apply its Weil factor to each $\phi_i(A)$. The resulting tensor product is multiplicative in A, and exact Egorov covariance preserves the CSS stabilizer line. Consequently the one-qudit Clifford scalar extension, restricted to the linear $\mathrm{SL}_2(q)$ factor, splits coherently with the propagated tensor symmetries. The exceptional uniqueness clause at $q = 3$ in Gérardin's theorem does not remove existence.

The affine extension does not split. Choose $a, b \in \mathbb{F}_q$ with $\chi(ab) \neq 1$, and put $X = X(a)$, $Z = Z(b)$. Their projective classes commute, whereas every choice of unitary representatives satisfies

$$ZX = \chi(ab)XZ, \qquad \chi(ab) \neq 1.$$

Scalar changes of X and Z do not change their commutator. Hence the translation subgroup $\mathbb{F}_q^2$ has no homomorphic lift to the one-qudit unitary Clifford group. The coherent odd-field lift is therefore $H_q \rtimes \mathrm{SL}_2(q)$, not a section of $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ through the scalar quotient. This is the Heisenberg central extension, not a residual metaplectic or Schur-multiplier obstruction on $\mathrm{SL}_2(q)$.

Neither conclusion decides the party-permutation extension in Corollary 3.6. Its quotient is the evaluation-set-dependent realized permutation group Π, and its kernel is the full fixed-party projective group Γ, rather than the scalar phases. Its splitting remains exactly the nonabelian factor-set triviality criterion stated there. □

For a stabilizer state, the hypotheses of Corollary 3.3 hold whenever every party belongs to a support S of size at least three whose full supported stabilizer label subgroup is an $\mathbb{F}_q$-linear plane and projects isomorphically onto the local Weyl-label plane at every party of S.

4 The non-GRS pencil and its quotient

Let H_t be the ordered matrix in (1.2), over a finite field of odd order, and put

$$G(t) = t^4 - 4t^3 + 7t^2 - 4t + 1.$$

The admitted non-GRS condition is

$$t(t-1)(t^2 - t + 1)(t^2 - 3t + 1)G(t) \neq 0. \tag{4.1}$$

Factoring the twenty 3-by-3 column minors gives the first four factors, which keep the arc and quotient coordinates nondegenerate. The determinant for a conic through the six columns is a unit times G, so $G = 0$ is exactly the GRS boundary. Write $C_t = \ker H_t$ and $\Psi_t = \Psi_{C_t}$, and let A, B, z be as in (1.3).

Theorem 4.1 (Projective and Clifford quotient)**.** For admitted t, u, the following are equivalent:

$$H_t \sim_{\mathrm{proj}} H_u, \qquad C_t \sim_{\mathrm{mon}} C_u, \qquad z(t) = z(u),$$

allowing arbitrary party permutations. If q is prime, these are also equivalent to $\Psi_t \sim_{\mathrm{LC}} \Psi_u$.

Proof. Set

$$y(t) = \frac{(t-1)^2}{t}.$$

Then $A = -4t^2y$, $B = t^2(y^2 - 1)$, and

$$z = \frac{(y - y^{-1})^2}{16}.$$

It follows that $z(t) = z(u)$ exactly when

$$y(u) \in \{y(t), -y(t), y(t)^{-1}, -y(t)^{-1}\}. \tag{4.2}$$

Each of the four relations is realized by a direct projectivity and a party permutation. In order, one may use

relation	projectivity
$y(u) = y(t)$	$\operatorname{diag}(1, (1-u)/(1-t), u/t)$
$y(u) = -y(t)$	$\operatorname{diag}(1, (1-u)/(1-t), -u/t)$
$y(u) = y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (u-1)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$
$y(u) = -y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (1-u)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}.$

The corresponding permutations are respectively the identity, $(5\,6)$, $(3\,5)(4\,6)$, and $(3\,5\,4\,6)$. All denominators and determinants are nonzero on the admitted locus.

Conversely, the ten signed products of complementary 3-by-3 brackets of H_t form the multiset

$$\{+A, +A, +A, -A, -A, -A, +B, +B, -B, -B\}. \tag{4.3}$$

Projectivity and independent column rescaling multiply all ten products by one common scalar; party permutation changes at most their common sign. On the admitted locus $A, B \neq 0$. If $A^2 \neq B^2$, the multiplicities three and two in (4.3) distinguish the two opposite pairs and recover $(B/A)^2 = z$. If $A^2 = B^2$, the multiset instead collapses to two opposite values, each of multiplicity five. That collapsed pattern is itself invariant under the common scaling and sign, and it is equivalent to $z = 1$. Thus the multiset recovers z in both cases, and projective equivalence forces equality of z. The equivalence with monomial codes follows from Section 2, and projective equivalence supplies an LC equivalence.

We now specialize the transition atlas from the rigidity argument to $m = 3$. Each four-party support identifies its two-dimensional stabilizer plane with the local Pauli-label plane at any retained party. Overlapping supports therefore give coordinate changes between local Weyl frames, and their compositions around a loop give holonomy up to local symplectic conjugacy. By (3.5), these minimum-support planes generate the full stabilizer label group. The 450 quantities below are conjugacy data of this atlas, not an independent choice of scalar contractions.

For the remaining direction assume that q is prime. Then a local Clifford induces a block $F_i \in \mathrm{SL}_2(q)$ on each local Pauli-label plane. Put $K_T = L_{C_t}(T)$ for every four-set T. If $i, j \in T$, projection $p_i : K_T \to \mathbb{F}_q^2$ onto the Pauli-label plane at i is an isomorphism by (3.7). Define the transition

$$M_T(i,j) = p_j p_i^{-1} : (\mathbb{F}_q^2)_i \longrightarrow (\mathbb{F}_q^2)_j$$

and, for distinct four-sets T, U containing i, j, the holonomy

$$\mathrm{Hol}(T, U; i, j) = M_U(j, i) M_T(i, j). \tag{4.4}$$

Local symplectic blocks F_i send this endomorphism to $F_i \,\mathrm{Hol}\, F_i^{-1}$. Reversing T, U takes its inverse, and a party permutation only relabels the data. Thus the multiset of

$$r(M) = \mathrm{Tr}(M)^2 / \det M$$

over the $15 \cdot 6 \cdot 5 = 450$ choices is an LC and party-permutation invariant.

Direct substitution of the shortened generators into (4.4) gives the following rational-function multiset; the symbolic identities and a direct-Lagrangian reconstruction are independently replayed by the paper-local certificate described in Section 7:

$$\begin{array}{c|c} 4 & 90 \\ -1/z & 144 \\ (2z+1)^2/z^2 & 24 \\ \text{each root of } X^2 - 8X + (8 - 16z - 1/z) & 96. \end{array} \tag{4.5}$$

Because $A, B \neq 0$ on the admitted locus, $z \neq 0$. The multiplicity 96 applies to each root before specialization; coincident values have their multiplicities added. The identity

$$4B(t)^2 + A(t)^2 = 4G(t)^2$$

shows that the excluded GRS locus is exactly $z = -1/4$. Away from it, the constant 4 bin is isolated and the two quadratic roots are distinct. The bin containing $-1/z$ can merge with neither, one, or both of the weight-24 and one weight-96 contributions, so its multiplicity is $144, 168, 240$, or 264. No other bin has one of these multiplicities. Hence the multiset canonically recovers $-1/z$, including every modular collision, and LC equivalence forces equality of z. $\square$

The map $t \mapsto z$ has degree eight: the four transformations in (4.2), followed by the quadratic relation $t+t^{-1} = y+2$, account for its eight sheets. The signed refinement $w = B/A$, $z = w^2$, is not quantum data. Both even and odd explicit party permutations reverse w, and their projectivities lift to monomial local Cliffords.

Proof of Corollary 1.5. Theorem 4.1 gives all equivalences except LU. Every LC is an LU, while Theorem 1.1 turns every LU intertwiner between the states Ψ_t, Ψ_u into an LC intertwiner. $\square$

Proposition 4.2 (Frobenius-sector divisors)**.** Let σ be an automorphism of a finite field of odd order and let t be admitted and non-GRS. Put

$$E_\sigma(t) = (\sigma(t) - t)(1 - \sigma(t)t), \qquad D_\sigma(t) = ((1 - \sigma(t))(1 - t))^2 + \sigma(t)t.$$

Then

$$\frac{\sigma(t)}{(1 - \sigma(t))^2} = \frac{t}{(1-t)^2} \quad \Longleftrightarrow \quad E_\sigma(t) = 0. \tag{4.6}$$

If $h = (1 - \sigma(t))(1 - t)$ and $w = (-h, -h, 1, 1, -1, -1)$, then

$$H_{\sigma(t)} \operatorname{diag}(w) H_t^{\mathsf{T}} = 0 \quad \Longleftrightarrow \quad D_\sigma(t) = 0. \tag{4.7}$$

Moreover $D_{\mathrm{id}}(t) = G(t)$, and

$$E_\sigma(t) = 0 \quad \Longrightarrow \quad D_\sigma(t) \neq 0. \tag{4.8}$$

Finally G, A, B, z commute with σ, and σ preserves the admitted non-GRS locus.

Proof. Admission gives $t \neq 1$, and injectivity of σ gives $\sigma(t) \neq 1$. Cross multiplication in (4.6) gives

$$s(1-t)^2 - t(1-s)^2 = (s-t)(1-st)$$

with $s = \sigma(t)$. Direct multiplication in (4.7) makes eight entries vanish identically and leaves $-2D_\sigma(t)$ in the last entry; odd characteristic gives the equivalence. Setting σ to the identity yields $D_{\mathrm{id}} = G$.

If $E_\sigma(t) = 0$, then either $\sigma(t) = t$ or $\sigma(t)t = 1$. In the first case $D_\sigma(t) = G(t)$. In the second, multiplying $D_\sigma(t)$ by t^2 gives $G(t)$. Admission excludes $t = 0$ and $G(t) = 0$, proving (4.8). The equivariance statements follow by applying σ to the defining polynomial and rational expressions. $\square$

The prime-field hypothesis is necessary for this scalar classification. Over $\mathbb{F}_{p^e}$, the basis permutation $|x\rangle \mapsto |x^p\rangle$ is a local Clifford and sends Ψ_t to Ψ_{t^p}, while $z(t^p) = z(t)^p$ need not equal $z(t)$. Concretely, in $\mathbb{F}_{25} = \mathbb{F}_5[s]/(s^2 - 2)$, both s and $-s = s^5$ are admitted, but

$$z(s) = 3 + 3s, \qquad z(-s) = 3 + 2s.$$

For each field automorphism σ, Proposition 4.2 gives the collision and diagonal-isodual divisors relating t to $\sigma(t)$. It does not show that an arbitrary additive local Clifford determines a nontrivial field automorphism, nor does it construct a local Clifford from a Galois match of z. Those two representation-theoretic bridges are not claimed here, so neither is a full extension-field LC/LU classification.

5 Diagonal isoduality and the logical-Clifford phase

Throughout this section q is an odd prime. Regard any one party of an AME$(6, q)$ stabilizer tensor as an input and the other five as outputs. This is a $[[5, 1, 3]]_q$ encoding isometry. In prime local dimension, the symplectic action of a one-qudit Clifford is a block in $\mathrm{SL}_2(q)$, so the kernel of the party action on Clifford tensor symmetries is

$$\mathcal{K}_C = \{(F_1, \ldots, F_6) \in \mathrm{SL}_2(q)^6 : (\bigoplus_i F_i)L_C = L_C\}.$$

For $q = p^e$ with $e > 1$, the full local Clifford group acts instead through $\mathrm{Sp}_{2e}(\mathbb{F}_p)$; the displayed field-linear block group is then not the full local Clifford kernel. For an encoder view, the *fixed-party logical symplectic image* means the image of $\mathcal{K}_C$ on its input block. Corollary 1.4 identifies the all-length phase boundary as diagonal isoduality. For six-arcs, classical self-association turns that intrinsic code condition into the conic criterion below.

Lemma 5.1 (Fixed-label self-association)**.** Let H be a rank-three 3×6 matrix whose columns are a six-arc, and put $C = \ker H$. There is a nonsingular diagonal matrix S with $SC = C^\perp$ if and only if the six columns of H lie on a conic.

Proof. Let G be a 3×6 generator matrix for C. Its columns are the labeled Gale transform of the columns of H. Since $C^\perp = \mathrm{row}(H)$,

$$SC = C^\perp \iff \mathrm{row}(GS) = \mathrm{row}(H) \iff AGS = H \text{ for some } A \in \mathrm{GL}_3(q).$$

Thus the condition is precisely fixed-label projective equivalence to the Gale transform. This fixed locus can be identified directly over the finite field. Let V be the matrix with quadratic evaluation rows

$$(x_i^2, y_i^2, z_i^2, x_iy_i, x_iz_i, y_iz_i).$$

Its determinant vanishes exactly when a nonzero quadratic form contains all six columns. Every five rows of V are independent: a relation supported on at most five points would give $H_IDH_I^T = 0$ for a nonsingular diagonal D on its support I; for $|I| \geq 3$, this would put the three-dimensional row space of H_ID inside $\ker H_I$, which has dimension at most two, while smaller supports are immediate. Hence any relation $\sum_i d_iV_i = 0$ has every $d_i \neq 0$.

Writing $D = \mathrm{diag}(d_i)$, that relation is $HDH^T = 0$. Therefore $\mathrm{row}(HD) \subseteq \ker H = C$, and equality follows from dimension three. Thus $CD^{-1} = C^\perp$. Conversely, a nonsingular diagonal equivalence $SC = C^{\perp}$ gives $HS^{-1}H^T = 0$, hence a relation among the rows of V and a conic through the six columns. The conic is irreducible because the columns form an arc. This is the finite-field fixed-label form of the classical six-point self-association theorem [HMSV08, Section 2.3]. $\square$

Theorem 5.2 (Fixed-party logical phase)**.** For an equal-phase CSS state from a six-arc over an odd prime field, the fixed-party logical symplectic image is

$$\begin{cases} \mathrm{SL}_2(q), & \text{if the arc is conic/GRS,} \\ T = \{\mathrm{diag}(a, a^{-1}) : a \in \mathbb{F}_q^{\times}\}, & \text{if the arc is nonconic.} \end{cases} \tag{5.1}$$

Thus every encoder view of a GRS tensor realizes the full one-qudit logical symplectic group from fixed-party symmetries. Off the GRS locus, the fixed-party logical symplectic image is the split torus. Any party-moving isoduality that exchanges the CSS X- and Z-spaces acts by the nontrivial Weyl-group element and generates the normalizer of T together with T.

Proof. Every four-party support carries the two-dimensional shortened label plane (3.7), and projection to the Pauli plane of any retained party is an isomorphism. If F_i is fixed at one party, requiring $\bigoplus_i F_i$ to carry each shortened plane to itself therefore determines the other five blocks through overlapping four-sets. These conditions are also sufficient: the shortened planes span L_C by the general minimum-support generation identity (3.5).

The common diagonal anchor $\mathrm{diag}(a, a^{-1})$ preserves $C_X \oplus C_Z^{\perp}$, so T is always present. Write a propagated collection as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix}, \qquad D_\alpha = \mathrm{diag}(\alpha_1, \dots, \alpha_6), \quad \text{and similarly for } D_\beta, D_\gamma, D_\delta.$$

Preservation of $L_C = C_X \oplus C_Z^{\perp}$ is equivalent to

$$\begin{aligned} D_\alpha C &\subseteq C, & D_\gamma C &\subseteq C^{\perp}, \\ D_\beta C^{\perp} &\subseteq C, & D_\delta C^{\perp} &\subseteq C^{\perp}. \end{aligned} \tag{5.2}$$

Apply Proposition 3.8 to $D_\gamma C \subseteq C^{\perp}$ and $D_\beta C^{\perp} \subseteq C$. A nondiagonal input block has a nonzero off-diagonal entry, hence the corresponding diagonal matrix is nonzero, nonsingular, and gives

$$D_\gamma C = C^{\perp} \qquad \text{or} \qquad D_\beta C^{\perp} = C. \tag{5.3}$$

Conversely, if a nonsingular diagonal S satisfies $SC = C^{\perp}$, then $S^{-1}C^{\perp} = C$. Taking $D_\gamma = S$, respectively $D_\beta = S^{-1}$, with identity diagonal blocks in (5.2) realizes the lower and upper elementary unipotent input anchors. Together with T, these generate $\mathrm{SL}_2(q)$.

Lemma 5.1 identifies the diagonal equivalence $C \sim C^{\perp}$ with the conic locus. On it, the standard GRS dual multipliers realize the code–dual equivalence and hence every anchor in $\mathrm{SL}_2(q)$; off it, the preceding implication excludes every nondiagonal anchor.

Finally, an isoduality interchanges the two summands C_X and $C_Z^{\perp}$, so its input block swaps the two Weyl axes. It conjugates a to a^{-1} in T, giving $N(T)$ whenever such a party-moving symmetry is present. $\square$

Corollary 5.3 (Exact fixed-party transversal group)**.** For the associated $[[5,1,3]]_q$ encoder, the projective logical Clifford group realized without moving parties is

$$\begin{cases} \mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q), & \text{if the six-arc is conic/GRS,} \\ \mathbb{F}_q^2 \rtimes T, & \text{if the six-arc is nonconic.} \end{cases}$$

Proof. Every logical Pauli has a physical Pauli representative, giving the normal translation subgroup $\mathbb{F}_q^2$. Theorem 5.2 identifies the exact linear factor realized by fixed-party tensor symmetries. Modulo scalar phases, the one-qudit Clifford group is the affine semidirect product $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$, so these two ingredients give the displayed equalities. □

This is an operational distinction, not a classical automorphism count. The fixed-party non-GRS group has order $q^2(q-1)$, while the GRS group has order $q^2|\mathrm{SL}_2(q)|$. Corollary 3.7 makes the party-moving comparison exact at $q = 11$: the two non-GRS pencil classes have party images S_5 and S_4, their extensions split, and odd party motion inverts T. Their party-moving logical group is therefore $\mathbb{F}_q^2 \rtimes N(T)$. The GRS and party-moving non-GRS logical-group orders are 159720 and 2420, respectively.

The MDS property also makes minimum operator pushing uniform. For any input party and nonidentity Pauli label, each of the ten four-sets containing that party gives a unique stabilizer representative supported there. Removing the input leaves a weight-three output Pauli. Thus each input Pauli has exactly ten minimum-support pushes, although the symmetry orbits of those pushes can distinguish finer projective classes. This uses only the projection isomorphism for the shortened planes, not the finite group calculation.

Arithmetic symmetry jumps do not change the phase. The characteristic-17 tetrahedral specialization lies on the GRS boundary and therefore retains the full $\mathrm{SL}_2(17)$ fixed-party image; its party image is S_4, and the computed extension splits. The characteristic-31 A_5 specialization is non-GRS, so its fixed-party image is the split torus; its S_5 party image also splits, and odd motion enlarges the logical linear group to $N(T)$. These are symmetry enhancements inside the two phases, not exceptions to (5.1).

6 Explicit local-unitary invariants

Theorem 1.1 settles equivalence without scalar invariants. Scalar witnesses remain useful because they are short, basis-free certificates and because their failures explain what the operator-valued marginal retains.

6.1 A marginal-moment separator

For a four-party set T, let ρ_T be the reduced state and embed it into all six parties as

$$A_T = \rho_T \otimes I_{T^c}.$$

For an unordered triple $\{T, U, V\}$ of distinct four-sets, the number $\mathrm{Tr}(A_T A_U A_V)$ is invariant under arbitrary product unitaries; party permutation merely permutes the 455 triples. By (2.3),

$$\mathrm{Tr}(A_T A_U A_V) = q^{-\operatorname{rank}(L_C(T)+L_C(U)+L_C(V))}. \tag{6.1}$$

Here and below, ranks of label spaces and matching systems are over $\mathbb{F}_q$. Indeed, the three stabilizer sums have total normalization q^{-12}. Only triples of labels with sum zero survive the full trace; isotropy of L_C removes multiplication phases, and the kernel of the addition map from the three two-planes has size q^{6-r}, where r is the $\mathbb{F}_q$-rank in (6.1).

Index a four-set by its omitted pair, an edge of K_6. The rank in (6.1) is four exactly when the three corresponding chord lines concur in both the six-arc $\mathcal{A}$ and its Gale dual $\mathcal{A}^*$, represented by any kernel basis of the arc matrix. To see this, split each shortened CSS plane into its one-dimensional $X(C)$ and $Z(C^\perp)$ parts. For three omitted edges, the three shortened X-lines span dimension two precisely when the corresponding chords of $\mathcal{A}^*$ concur; the Z-lines give the same criterion for $\mathcal{A}$. Each span has dimension at least two, so their sum has rank four exactly when both concurrences hold.

Sixty triples are stars and concur universally. Any other three-edge graph that contains two adjacent edges but is not a star has two chords meeting at their common arc point and a third chord missing it, since the columns form an arc. The only remaining graph type is a perfect matching. Hence

$$\#\{q^{-4} \text{ moments}\} = 60 + b(\mathcal{A}, \mathcal{A}^*), \tag{6.2}$$

where b counts perfect matchings concurrent in both configurations.

Lemma 6.1 (Concurrent matchings on a conic)**.** Let S be six distinct points on a nonsingular conic over a finite field k of odd characteristic other than 5. At most six perfect matchings of S have concurrent joining chords.

Proof. Identify the conic with $\mathbb{P}^1$. Projection from a point off the conic pairs the points in every fiber and hence induces a projective involution; conversely, the chords joining the three two-element orbits of a projective involution meet at its center. Thus the concurrent matchings are exactly the fixed-point-free involutions in

$$H = \mathrm{Stab}_{\mathrm{PGL}_2(k)}(S).$$

The correspondence is injective because an element of $\mathrm{PGL}_2(k)$ is determined by its action on three points. Moreover, H acts freely on ordered triples of distinct points of S, so $|H| \mid 6 \cdot 5 \cdot 4 = 120$.

Dickson's classification, including the p-regular cases [Dic01, §§239–261], and Faber's p-irregular classification [Fab23, Theorem 1.1] now leave the following possibilities. The last column bounds involutions having no fixed point on this six-set:

H	fixed-point-free involutions
cyclic	≤ 1
dihedral	≤ 6
A_4	≤ 3
S_4	≤ 6
A_5 in its faithful degree-six action	0
C_3 or S_3 (char $k = 3$)	≤ 3.

For the A_5 row, each involution fixes two of the six points. In the dihedral row, if the rotation order exceeded six, every orbit away from the two rotation-fixed points would have size at least that order, so no invariant six-set could exist; for rotation order at most six the two reflection classes together contribute at most six fixed-point-free elements (and the possible central half-turn contributes only when one reflection class has fixed points). In the characteristic-three semi-elementary case, the normal 3-group has order 3, while its cyclic complement embeds in $\mathrm{Aut}(C_3)$, giving only C_3 or S_3. The subfield groups whose order divides 120 are $\mathrm{PSL}_2(3) \cong A_4$ and $\mathrm{PGL}_2(3) \cong S_4$, already listed. The remaining characteristic-five subfield possibilities are excluded by the hypothesis. Every row is therefore bounded by six. $\square$

Theorem 6.2 (Uniform H_3/GRS separation)**.** Let $\tau^2 = \tau + 1$, and let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\tau]$ of odd residue characteristic for which the reduction of the integral H_3 (icosahedral) sextuple

$$((0, 1, 1-\tau), (0, 1, \tau-1), (1, 1-\tau, 0), (1, \tau-1, 0), (1, 0, -\tau), (1, 0, \tau))$$

is a six-arc. Write $k = \mathbb{Z}[\tau]/\mathfrak{p}$. If the reduced arc is non-GRS, its equal-phase state is not LU-equivalent, even after party permutation, to any six-point GRS state over k. (The hypothesis is vacuous in characteristic five, where the reduction is itself GRS.)

Proof. Exact integral chord determinants give ten common concurrent matchings for the H_3 arc and its Gale dual. The other five determinant norms are -64 and -4, so the count is exactly ten in every residue reduction covered by the theorem. This is an exact fifteen-matching calculation in $\mathbb{Z}[\tau]$, not a field sample. The H_3 moment multiset therefore contains 70 copies of q^{-4}.

Lemma 6.1 gives at most six concurrent matchings on a conic. The bound is attained whenever $\mathrm{PGL}_2(q)$ contains an octahedral S_4, by the corresponding six-set: the six half-turns about axes through pairs of opposite edges act without fixed vertices. Thus a GRS state has at most 66 copies of q^{-4}, and the margin $70 > 66$ is sharp for this argument. In characteristic five no subgroup bound is needed: $X^2 - X - 1 = (X-3)^2$, so the residue of τ is 3, and direct substitution gives $G(\tau) = 0$. The H_3 reduction is therefore GRS, contrary to the theorem's non-GRS hypothesis. Equation (6.1) is LU-invariant, so the gap proves the claim. □

The five nonconcurrent H_3 matchings form a one-factorization of K_6. The marginal incidence recovers its S_5 stabilizer but forgets the even A_5 orientation. This is necessary: full local Fourier transform sends Ψ_C to $\Psi_{C^\perp}$, and an integral isoduality returns the H_3 code through an odd pentad permutation in every odd reduction. The orientation is an LC-forgettable CSS electric/magnetic marking, not a separate LU class.

6.2 The exact dimension-thirteen four-copy witness

For r ket copies and r bra copies, choose a permutation $\sigma_i \in S_r$ at each party and contract ket copy a with bra copy $\sigma_i(a)$. If G generates a three-dimensional code, let $M_{\boldsymbol{\sigma}}(G)$ be the homogeneous matching system

$$(u_a G)_i = (v_{\sigma_i(a)} G)_i \quad (1 \le i \le 6,\ 1 \le a \le r).$$

Counting its solutions gives the normalized contraction

$$I_{\boldsymbol{\sigma}}(\Psi_C) = q^{3r - \operatorname{rank} M_{\boldsymbol{\sigma}}(G)}. \tag{6.3}$$

Theorem 6.3 (Exact four-copy separator at $q = 13$)**.** The two admitted $q = 13$ pencil classes with $z = 4$ and $z = 12$ are not local-unitary equivalent, even after arbitrary party permutation.

Proof. At $q = 13$, the admitted pencil has two classes in the same complete three-copy and marginal-moment bucket, with $z = 4$ and $z = 12$. Use the explicit four-copy pattern

$$\boldsymbol{\sigma} = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

For $\pi \in S_6$, put $(\pi\boldsymbol{\sigma})_i = \sigma_{\pi^{-1}(i)}$, and sum (6.3) over all party permutations:

$$J_{\boldsymbol{\sigma}}(\Psi) = \sum_{\pi \in S_6} I_{\pi\boldsymbol{\sigma}}(\Psi).$$

Over $\mathbb{F}_{13}$, the $z = 4$ class has rank 21 for all 720 terms. The $z = 12$ class has rank 20 for 192 terms and rank 21 for 528. Therefore

$$J_{\boldsymbol{\sigma}}(z=4) = 720 \cdot 13^{-9}, \qquad J_{\boldsymbol{\sigma}}(z=12) = 3024 \cdot 13^{-9}. \tag{6.4}$$

This proves arbitrary-LU inequivalence directly. The complete two- and three-copy contraction sets do not separate this pair; (6.4) makes no claim that four copies are globally minimal. □

6.3 Why fixed-copy scalars cannot recover the pencil coordinate

Proposition 6.4 (Generic constancy)**.** Fix r and a field k of cardinality q, which is also the local Hilbert-space dimension under the code–state dictionary. Fix an irreducible reduced parameter variety X over k, and a regular generator-matrix chart $G(x)$ of constant code dimension. Every degree-(r, r) permutation contraction of the associated equal-phase states has one value at every k-point of some dense open subscheme $U \subseteq X$ defined over k. The same holds for party-orbit sums and linear combinations. After evaluating the states as vectors in $(\mathbb{C}^q)^{\otimes n}$, in the stable range $q \geq r$, every scalar LU-invariant of bidegree (r, r) is generically constant in this sense. For a reducible family, the statement holds separately on each irreducible component.

Proof. Formula (6.3) depends only on the rank of a matrix whose entries are regular functions over k. A maximal nonzero minor shows that this rank is constant on a dense open subscheme defined over k. There are only finitely many diagrams at fixed r, so their generic-rank opens have a common dense intersection because X is irreducible. Over the complex vector space $(\mathbb{C}^q)^{\otimes n}$, applied pointwise to each code state, the first fundamental theorem for unitary invariants says that local copy-permutation diagrams span the bidegree-(r, r) invariants; when the local Hilbert-space dimension q is at least r, there are no additional dimension relations among these diagrams. This proves the last statement. $\square$

The dense open here may have no k-point over a small finite field; in that event its assertion about k-points is vacuous. The rank statement over the geometric generic point remains nonvacuous, while an application to a particular finite field must separately exhibit a point of $U(k)$.

For the pencil viewed over its rational-function field, the generic four-copy contraction values therefore generate only the constant field, not $\mathbb{Q}(z)$. The witness (6.4) detects a rank-jump stratum. In contrast, the four-party operator tensor (3.2) retains the Weyl axes and forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

7 Verification, literature, and scope

The rigidity theorem, the arc–AME dictionary, and generic constancy are conceptual proofs in the text. The projective quotient uses displayed bracket identities and projectivities. Its transition-map invariance and collision argument are proved in Section 4, while the 450 exact rational functions are certificate-supported. Section 3 proves the general MDS diagonal-multiplier criterion. Section 5 combines it with Gale duality to obtain the six-arc phase theorem. The party groups, factor sets, complements, and torus actions in Corollary 3.7 are certificate-supported finite examples, not inputs to the uniform theorem. Section 6 proves the marginal trace/rank and incidence bridges; the H_3 determinants and $q = 13$ contractions are certificate-supported. Appendix A proves the systematic transport reduction; the cycle-cover determinants, assignment exhaustion, exact minors, and double-coset support sets are certificate-supported.

The reproducibility package is in `supplement/`. From that directory,

`python3 verify.py`	checks all seventeen byte counts and SHA-256 digests,
`python3 verify.py --replay`	regenerates all eight canonical JSON certificates in memory.

The full replay uses only the Python standard library, deterministic integer, rational, polynomial, and finite-field arithmetic, and canonical JSON serialization. It includes separately implemented checks appropriate to each claim: CSS versus direct-Lagrangian shortening, symbolic versus finite projectivity, chord concurrency versus stabilizer ranks, quotient minors versus finite-field elimination, and cycle-cover versus fraction-free transport determinants. These checks belong to one paper-owned Python package; they are not external independent reproductions. The exact files, hashes, domains, expected outputs, and negative scopes are recorded in `supplement/EVIDENCE.md` and `supplement/EVIDENCE-MANIFEST.json`. On the reference machine the digest check takes under a second, the full evidence replay about six minutes, and each Lean aggregate under ten seconds.

From the repository root, the exact formal replay is

```
python3 lean/scripts/lean-build-queue.py run \
  RelativeConicArcs.Gates.AMELUAggregate \
  RelativeConicArcs.Gates.AMELUAggregateAxioms \
  --profile single --threads 1 \
  --aggregate RelativeConicArcs.Gates.AMELUAggregateAxioms
```

The checked release is then reproduced from `papers/ame_lu/` by

```
make release-check
```

The release identifier is content based: `release/RELEASE-MANIFEST.json` records the SHA-256 digest of every public byte and every file in the recursive formal dependency closure, including the Lean toolchain and build definitions. The manifest, rather than a mutable working-tree name, is the immutable identity of the refereed local release; a public DOI or archive URL must be assigned by the author at deposit.

Claim family	Proof mode	Machine trust / formal status
LU-to-LC rigidity, atlas classification, and no-go	manuscript proof plus Lean	support, transition, marginal, covariance, axis, and holonomy cores kernel checked; exact projective atlas extension proved in the manuscript
Pencil and six-arc classification	proof plus exact certificates	conditional Lean interface; geometric and holonomy inputs named
H_3 and $q = 13$ separators	proof plus exact certificates	finite graph core unconditional; trace/covariance and exact ranks named
Diagonal-isodual logical groups	manuscript proof plus Lean	multiplier theorem unconditional; exact carrier interface conditional
Twelve party splittings and transport	exact certificates plus abstract Lean	splitting algebra unconditional after a supplied complement; finite witnesses and transport matrices certificate supplied

No module in the aggregate uses `sorry` or unsafe declarations. Native execution is confined to three finite graph cardinalities proved by `native_decide`; the axiom terminal reports them separately. It prints declaration-level dependencies: the ordinary kernel-checked results use only `propext`, `Classical.choice`, and `Quot.sound`; all other finite evidence used by the paper is replayed by the Python supplement rather than imported as a Lean axiom.

The trusted boundary is Python integer arithmetic, the exact-arithmetic implementations in the generators, deterministic Gaussian elimination, and the mathematical dictionaries connecting arcs, codes, stabilizers, and tensor contractions. The certificates check the bounded enumerations and symbolic identities that they record. They do not prove the conceptual correspondence between the implemented objects and the paper definitions; that correspondence is supplied in the preceding sections.

The accompanying Lean 4 artifact uses toolchain `v4.32.0-rc1`. Its aggregate terminal `RelativeConicArcs.Gates.AMELUAggregate` checks the complete arc–MDS–CSS–AME dictionary, the algebraic pencil quotient, the finite marginal graph counts, the contraction normalization and party-orbit identities, the length-generic MDS–CSS LU-to-LC chain, the arbitrary-additive support squeeze, minimum-support generation, realized full-Weyl marginal chain, LU-to-LC terminal, linear minimum-support transitions, and atlas-equivalence relation, the abstract holonomy-centralizer mechanism for both normal and arbitrary local-block subgroups, the extension-field pencil-sector algebra, and the transport polynomial and characteristic-seven identities in one environment. The arbitrary-additive layer defines coordinate restriction of a prime-field label space, identifies its kernel with supported labels, and proves that absence of a nonzero label on at most m parties makes the one-site kernel projection injective. The exact $2m$ versus $m-1$ dimension balance then makes that projection bijective and gives a unique supported label over every local Weyl label. A separate support-profile interface proves $L(A) = L(A \setminus \{i\}) + L(A \setminus \{j\})$ and the resulting generation by $(m+1)$-supported subspaces. The diagonal-tensor modules prove the arity-independent axis lemma for arbitrary bijective local reindexings and nonzero coefficients. The abstract holonomy module proves that compatible local gauges are the centralizer of loop transports. Its arbitrary-subgroup form retains the exact propagated-membership conditions needed for $\mathrm{Sp}_{2e}(\mathbb{F}_p)$; its normal-subgroup specialization gives the pure $\mathrm{SL}_2(q)$ centralizer in prime dimension.

The `StabilizerAMERigidity` module composes the arbitrary-additive support profile, unique local-label recovery, phased full-Weyl marginal, marginal covariance, and axis recovery into fixed-party and party-relabeled LU-to-LC terminals. Its sole physics-facing realization field is the standard partial-trace stabilizer-projector expansion (2.2); the party-relabeled terminal also takes an explicit realization of the permuted source state. Constructing those fields from a lower-level Pauli representation remains outside this module, but no further rigidity step is assumed. The earlier MDS–CSS specialization remains unconditional in the generic Lean chain, including marginal covariance, retained-set coverage, and party relabeling. Composition with the admitted-pencil interface gives the formal conditional statement LU $\iff z(t) = z(u)$ in that specialization. The same artifact constructs the one-site Clifford quotient by nonzero scalars, proves the scalar subgroup closed, and proves the quotient Hausdorff, discrete, and finite by recording exact Weyl conjugates and constraining their scalar factors through determinant polynomials. It embeds the product-automorphism quotient into a finite product of these one-site quotients. The exact adjoint-signature map is continuous into a finite Hausdorff space, while its identity fiber is the connected product of one-site unit circles. This proves the scalar-phase identity-component statement, both with fixed party labels and after adjoining the discrete party-permutation factor. The scalar inclusions are continuous homomorphisms, every connected component is a scalar-torus translate, and finitely many such components cover either automorphism group. The party-permutation extension has a section-free outer action and a normalized nonabelian factor set. Its ordered change-of-section law makes trivializability independent of the chosen lifts, and trivializability is equivalent to a homomorphic splitting. The companion `PartyExtensionSplitting` module proves that a complement supplies the trivializing cochain, semidirect-product equivalence, unique kernel–quotient coordinates, cardinality product, and the involutive party-extension lift whose conjugation action on T is inversion. This projective involution is distinct from the determinant-one matrix J, which satisfies $J^2 = -I$. The twelve concrete complements in Corollary 3.7 are exact certificate computations rather than kernel-checked finite tables.

The geometric classification, marginal-trace, logical propagation, finite contraction, diagonal-isodual group, and transport orbit steps enter separate hypothesis structures; the paper constructs or certificate-supports them as specified above, but the Lean modules do not. Importing the terminal therefore does not make them unconditional formal theorems. The Choi inverse-transpose bridge, Clifford transpose closure, and factorwise transversal no-go are kernel checked in `EncoderTransversal`. The companion `DiagonalIsoduality` module proves the arbitrary-length MDS diagonal-multiplier lemma, the zero-or-one-dimensional multiplier space, the exact nullity test, and projective uniqueness of the witness ratios. The `StabilizerDictionary` module proves that the Pauli symplectic pairing identifies the ambient label space with its dual and that every linear stabilizer-character exponent is induced by an ambient Pauli label; this is the phase-correction step used in Section 3. The exact diagonal-isodual transversal-group dichotomy remains conditional: special-linearity, torus propagation, diagonal-duality propagation, the off-diagonal block-to-multiplier bridge, and the complete translation fiber remain named fields. Section 3 proves these action-level inputs for arbitrary $[2m, m, m+1]$ MDS codes. The fixed-copy generic-constancy proposition, the party-moving logical normalizer, and the determinant and double-coset inputs of the complete transport theorem are not formalized. The companion `AMELUAggregateAxioms` terminal separates the three native finite graph counts from the ordinary kernel-checked declarations. For each conditional structure, the statement-adequacy ledger records its fields individually. Some fields are constructed in the displayed paper proof, such as the four deck projectivities and the marginal trace/rank reduction; some are mathematical implications supported by a paper-local certificate, such as the H_3 determinant count and the four-copy rank evaluations; and some combine a prose construction with a certificate replay, such as the transport determinant and double-coset inputs. These are hypotheses of formalized implications, not additional Lean theorems.

The extension-field scalar statements in Proposition 4.2 are unconditional in Lean. The aggregate audit checks the frame-ratio factorization, explicit Gale multiplier and odd-characteristic zero criterion, same-exponent sector disjointness, and automorphism equivariance of G, A, B, z and the admitted locus. Its separate Galois-z orbit terminal is conditional on two named bridges: extraction of a projective Frobenius sector from an additive Clifford equivalence, and construction of a Clifford from a Galois match. Neither bridge is used as a paper theorem.

The LU-to-LC mechanism has direct qubit ancestry. Rains recovered the three Pauli axes of distance-two code projectors [Rai99]; Van den Nest, Dehaene, and De Moor used minimal stabilizer supports to force every local factor of an LU equivalence to be Clifford [VdNDDM05b]. Nonstabilizer minimal-support AME states permit different behavior [BR20, RL25]. The principal contribution here is the arbitrary-additive, all-prime-power stabilizer-AME theorem. Its MDS–CSS specialization makes the exact odd-prime fixed-party dichotomy computable: $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ on the diagonally isodual branch and $\mathbb{F}_q^2 \rtimes T$ otherwise, with the GRS tower on the first branch.

Finally, none of the results classifies nonlinear orthogonal arrays, arbitrary nonstabilizer AME tensors, or stabilizer states that are not AME. We do not claim that four copies are globally minimal, that $z = 2, 4/9$ are orbit exceptions, or that a fixed copy degree separates all classes as q varies. The logical-group theorem and the quantum part of the pencil classification are restricted to odd prime fields. Over extension fields, Frobenius basis permutations are additional local Cliffords. The scalar z is Frobenius-covariant, and Proposition 4.2 determines the two divisors attached to each field automorphism, but the two additive-Clifford bridges needed for a full orbit classification are outside the theorem package. The minimum-support atlas theorem remains exact over extension fields, with local blocks in $\mathrm{Sp}_{2e}(\mathbb{F}_p)$; its holonomy form requires both a centralizing base block and the propagated symplectic membership conditions, and does not imply a semilinear classification. Theorem 6.2 is restricted to the stated odd residue-field reductions, and the detector exceptions described in Appendix A do not alter the characteristic-uniform rigidity theorem.

The central distinction is therefore between covariant and scalar data. A four-party marginal retains the full configuration of local Weyl axes, and that configuration rigidifies every product-unitary intertwiner. Fixed-copy scalar contractions discard those axes and are generically constant, although their rank-jump strata still reveal exact divisors. For the admitted pencil, the covariant rigidity returns the classical degree-eight quotient z as the complete LU coordinate.

A Transport calculation for the four-copy witness

We now explain the rank jump behind (6.4). Fix the six copy permutations

$$\boldsymbol{\sigma} = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

Let V be the three-dimensional message space and let $g_i : V \to k$ be the six coordinate covectors. Form a bipartite four-sheeted cover with vertices L_a, R_b, $0 \le a, b < 4$, and a color-i edge from L_a to $R_{\sigma_i(a)}$. Put V at each vertex and k at each edge, using $g_{p(i)}$ as both restriction maps for a party assignment $p \in S_6$. A global section is a tuple $(x_a, y_b) \in V^8$ satisfying

$$g_{p(i)}(x_a - y_{\sigma_i(a)}) = 0 \tag{A.1}$$

on all 24 edges. The three-dimensional constant-section space is always present. Gauging it away leaves the 24-by-21 matching matrix in (6.3).

Let $U = k^4/k(1, 1, 1, 1)$, and let ρ_i be the action of σ_i on U. For each p, the MDS property makes $g_{p(3)}, g_{p(4)}, g_{p(5)}$ a basis of V^*. Define Q_p by

$$g_{p(i)} = \sum_{k=0}^{2} (Q_p)_{ik} g_{p(3+k)} \qquad (0 \le i < 3). \tag{A.2}$$

Thus Q_p is the left block obtained by systematizing the party-reordered generator matrix on its last three columns; no pivot can vanish because every three coordinate covectors are independent. Put $\widetilde{Q}_p = 2(t-1)Q_p$. This scalar is a unit on the admitted locus. For the identity assignment, the polynomial representative is

$$\widetilde{Q}_1 = \begin{pmatrix} a & b & c \\ a & c & b \\ d & d & d \end{pmatrix}, \qquad \begin{array}{ll} a = -2(t-1)^2, & b = -(t^2 - t + 1), \\ c = -(t^2 - 3t + 1), & d = -2(t-1). \end{array}$$

If $Y_k \in U$ are the three free copy vectors, then (A.1) reduces to the 9-by-9 block transport operator

$$\widetilde{T}_p = \big((\widetilde{Q}_p)_{ik}(\rho_i - \rho_{3+k})\big)_{i,k=0}^{2}. \tag{A.3}$$

The dimension identity

$$21 - \operatorname{rank} M_{\pi\boldsymbol{\sigma}}(G(t)) = 9 - \operatorname{rank} \widetilde{T}_p \tag{A.4}$$

holds over the field k, where $p = \pi$ in the convention of Section 6. It follows by quotienting the three-dimensional constant-section kernel and then eliminating the three free systematic coordinates. Multiplying the resulting operator by the unit $2(t-1)$ does not change its rank. All steps are invertible on the admitted six-arc locus.

Theorem A.1 (Transport divisor)**.** For a parameter on the admitted odd non-GRS pencil, there exists a party assignment $p \in S_6$ for which the sheaf (A.1) has a nonconstant section if and only if

$$(B^2 - 2A^2)(9B^2 - 4A^2) = 0,$$

or equivalently

$$(z-2)(9z-4) = 0. \tag{A.5}$$

At a generic point above $z = 2$, exactly 96 party assignments have rank 20; at a generic point above $z = 4/9$, exactly 192 do. Every other assignment has rank 21.

Proof. For the identity systematic chart, the coefficients satisfy

$$a = b + c, \qquad d^2 = -2a, \qquad A = a(c - b), \qquad B = bc.$$

Common copy relabelling, a change of basis in V, and rescaling a coordinate covector act on $\widetilde{T}_p$ by invertible row and column operations. Modulo these operations, the 720 party assignments split into boundary types and three nonboundary relative-cover types. Here the type records the six relative copy permutations $\rho_j^{-1}\rho_i$, equivalently the even-cycle pattern in each pair of colored matchings. Expanding a representative determinant by signed cycle covers gives

$$\begin{aligned} P_- &= AB(3B - 2A), \\ P_+ &= AB(3B + 2A), \\ P_2 &= A(B^2 - 2A^2). \end{aligned}$$

With integral bases the determinants are respectively

$$64d^3AB(3B - 2A), \quad 64d^3AB(3B + 2A), \quad 64d^3A(B^2 - 2A^2).$$

All omitted factors are units on the admitted odd pencil. Every assignment outside these three types has a determinant that is a unit there. The signed cycle-cover ledger, the exhaustion of the 720 assignments, and a fraction-free determinant path are recorded in the paper-local certificate. The two signed types lie above $w = B/A = \pm 2/3$, hence $z = 4/9$; the axial type lies above $z = 2$. Fraction-free elimination also gives a nonzero 8-minor on each component, so the generic kernel is exactly one-dimensional.

The row/column equivalences organize the exceptional assignments into double cosets. The axial stabilizers have orders 48 and 16, with intersection of order 8, so its support has size

$$48 \cdot 16/8 = 96.$$

For either signed type the corresponding orders are 24, 48, and 6, giving

$$24 \cdot 48/6 = 192.$$

The two signed types are distinguished by the sign of w, while the axial cycle pattern differs from both; hence their support sets are disjoint. Explicit representatives, stabilizer intersections, and complete support sets are in the certificate. Equation (A.4) turns the one-dimensional transport kernels into rank 20 of the matching matrix. $\square$

The exceptional arithmetic belongs to this detector, not to Theorem 1.1. The identity $4B^2 + A^2 = 4G^2$ gives, respectively,

char k	specialized identities
3	$B^2 - 2A^2 = G^2, \quad 9B^2 - 4A^2 = -A^2,$
5	$9B^2 - 4A^2 = 4G^2,$
7	$9B^2 - 4A^2 = 2(B^2 - 2A^2).$

Thus in characteristic 3 both admitted components disappear; in characteristic 5 the $z = 4/9$ component lies on the excluded GRS boundary; and in characteristic 7 the two components merge. The disjoint 96- and 192-element supports then give the exact histogram $20^{288}21^{432}$. Characteristics $11, 13, 41$ introduce only pullback ramification, with no further rank drop. These reductions and the nonzero 8-minors that control them are replayed in the certificate.

The divisor records where one scalar contraction changes rank. It is not an LU-orbit boundary and, by Proposition 6.4, cannot be promoted to a generic coordinate.

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